

Boundary Completeness from Failed Gluing

Missing interface variables, predictive completion, and an edge-mode toy model

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Abstract

This paper studies a diagnostic problem for composable region weights. A region weight assigns a total weight to interior histories compatible with specified boundary data. If a region is split along an auxiliary interface, composition should be recoverable by summing or integrating over complete interface data. The question addressed here is what can be inferred when a proposed coarse interface variable fails that gluing test. In the finite core setting, an exact interface variable may be replaced by a coarser candidate. The exact gluing error is a weighted sum of conditional covariances between the left and right subregion factors, computed inside the hidden fibers discarded by the coarse variable. The identity itself is a finite law-of-total-covariance calculation; its use here is as a local diagnostic for missing boundary data. The paper then defines the factor-generated predictive completion, meaning the coarsest refinement of the proposed boundary on which the admissible factors needed for gluing are already well-defined. In sharp large-deviation regimes, omitting the completion variable becomes a synchronization gap: the positive difference between optimizing a hidden interface variable jointly across the cut and optimizing it independently on the two sides. Finite-memory stochastic systems and a finite gauge-frame toy model illustrate the mechanism. The gauge-frame example gives a controlled finite analogue of the edge-mode lesson: boundary variables that look redundant before cutting may be required for local composition after cutting.

1 introduction

Many formulations of physics assign data to regions and then require that larger regions be recoverable by composing smaller ones. Transition kernels compose by summing over intermediate states. Path integrals compose by integrating over fields on an internal hypersurface [2]. General-boundary approaches attach state spaces or amplitudes to arbitrary boundaries [12]. Open Markov processes and categorical approaches make similar compositional requirements explicit [7]. In gauge theory and gravity, cutting a system into subregions introduces boundary subtleties: local subsystems may require additional boundary variables or edge modes in order to compose correctly [3, 4, 6, 5].

This paper asks a narrow inverse question. Suppose a theory has a complete interface variable through which region weights compose. Suppose one proposes to replace that interface variable by a coarser one. What does failure of gluing over the coarse variable diagnose?

The answer developed here is finite and elementary, but it is useful because it separates three issues that are often mixed together. A failed gluing test can mean that the proposed boundary variable is incomplete, that the proposed cut is not admissible, or that the chosen region weight is not the object that composes. This paper focuses on the first case. If the underlying cut and weight are fixed and known to compose through a finer interface, then failure after coarse-graining has a precise obstruction: conditional dependence inside the hidden fiber.

The novelty is not the covariance identity by itself. The novelty is its use as a boundary-completion diagnostic for composable region weights. In standard statistics, sufficient statistics compress data while preserving information about a specified inferential target [13, 14, 15, 18]. In finite Markov-chain theory, lumpability asks when a projection of a Markov chain remains Markov [16]. Predictive-state representations encode dynamical state by predictions of future observations [17]. The construction here is related in spirit but has a different target: it asks which interface variables are sufficient for gluing local region weights across an auxiliary cut. In relation to Fritz’s categorical treatment of sufficient statistics, the completion here is not a general categorical sufficiency theorem; it is a finite gluing-specific analogue in which the admissible factors play the role of the tests preserved by the quotient. A companion paper develops the direct reconstruction problem from boundary responses; the present paper addresses the inverse diagnostic problem of detecting missing boundary-completion variables [1].

The contribution is fivefold.

1. The paper proves a covariance obstruction formula. When a complete interface $y \in \mathcal{B}$ is projected to a coarse candidate $c = \pi(y)$, the difference between complete gluing and coarse product gluing is the integral of the conditional covariance of the left and right factors over the fibers of π .
2. It defines a factor-generated predictive completion. Relative to a chosen admissible class of left and right factors, the completion is the coarsest refinement of the candidate boundary through which all admissible factors descend. This makes “missing boundary variable” a quotient statement rather than a metaphor.
3. It gives a constructive finite completion algorithm. For finite admissible factor families, the completion is computed by grouping complete interface values with the same coarse label and the same admissible left-factor and right-factor values.
4. It records the corresponding sharp-limit diagnostic. If a hidden completion variable is optimized away separately on each side of a cut, the composed sharp generator is generally too small. The nonnegative synchronization gap measures the missing common optimizer condition.
5. It gives two examples. A finite-memory stochastic process shows that a boundary must retain the memory variables on which future transitions genuinely depend. A finite gauge-frame model shows that quotienting out a shared frame before gluing loses correlations that local composition needs.

This is also the scope boundary. The paper does not derive continuum Yang-Mills or gravitational edge modes. The finite gauge-frame model is only a controlled test case. Its role is to show why edge-mode variables are natural from a compositional point of view: variables that are redundant in an uncut description may become necessary when two local descriptions must be glued.

The paper is organized as follows. Section 2 defines region weights, interface factors, and coarse boundary proposals. Section 3 proves the covariance obstruction. Section 4 defines factor-generated completion and proves its finite minimality property. Section 5 gives the sharp-limit synchronization gap. Sections 6 and 7 work through the finite-memory and finite gauge-frame examples. Section 8 relates the result to edge modes and states the limitations of the finite analysis. Section 9 concludes.

2 region weights and coarse interface proposals

The formalism begins with a complete interface model. The word “complete” is relative to a specified class of regions, cuts, and local factors; it is not an absolute property of a variable. The main results

in this paper are finite. The standard-Borel analogue requires regular conditional probabilities and measurable quotient statements; it is described only as an extension direction in Remark 3. The admissible factor families below are model inputs: they represent the chosen class of left and right region pieces whose gluing is being tested.

Definition 1 (complete finite interface model). *Let $(\mathcal{B}, \mathcal{F}, \mu)$ be a finite probability space. Let $\mathcal{L}$ be a family of nonnegative left factors $L : \mathcal{B} \rightarrow [0, \infty)$, and let $\mathcal{R}$ be a family of nonnegative right factors $R : \mathcal{B} \rightarrow [0, \infty)$. The complete gluing pairing is*

$$\langle L, R \rangle_{\mathcal{B}} := \sum_{y \in \mathcal{B}} L(y) R(y) \mu(y).$$

The interface variable $y \in \mathcal{B}$ is complete for $(\mathcal{L}, \mathcal{R})$ when, for every admissible pair (L, R) , this pairing is the region weight assigned by the model to the composed region.

The abstraction hides the external boundary values. In an actual region decomposition, L is obtained from a left subregion weight after fixing its external boundary data, and R is obtained similarly on the right. The interface obstruction is already visible in the resulting functions of the shared interface value y .

Definition 2 (coarse interface proposal). *A coarse interface proposal is a map*

$$\pi : \mathcal{B} \rightarrow \mathcal{C}, \quad c = \pi(y),$$

where $\mathcal{C}$ is the proposed boundary data space. Let $\nu = \pi_ \mu$, so*

$$\nu(c) = \sum_{y \in \pi^{-1}(c)} \mu(y).$$

For every c with $\nu(c) > 0$, let μ_c be the conditional probability on the fiber $\pi^{-1}(c)$:

$$\mu_c(y) = \frac{\mu(y)}{\nu(c)} \quad \text{for } y \in \pi^{-1}(c).$$

For any factor $F : \mathcal{B} \rightarrow [0, \infty)$, write

$$\bar{F}(c) := \sum_{y \in \pi^{-1}(c)} F(y) \mu_c(y)$$

for its conditional average over the fiber.

The coarse proposal stores only c . The most direct coarse product gluing uses the averaged left and right factors:

$$\langle L, R \rangle_{\mathcal{C}}^{\text{prod}} := \sum_{c \in \mathcal{C}} \bar{L}(c) \bar{R}(c) \nu(c).$$

The covariance theorem below measures the difference between this coarse rule and the complete pairing. As an algebraic identity it only requires the finite space, projection, measure, and factor families. Completeness supplies the interpretation that the finer interface already gives the correct region weight. Proposition 1 shows that the failure is exactly the conditional covariance left inside the hidden fibers.

Definition 3 (pairwise and robust coarse gluing). *For a fixed pair (L, R) , pairwise coarse gluing succeeds when*

$$\langle L, R \rangle_{\mathcal{B}} = \langle L, R \rangle_{\mathcal{C}}^{\text{prod}}.$$

Robust coarse gluing succeeds for the class $(\mathcal{L}, \mathcal{R})$ when

$$\text{Cov}_{\mu_c}(L, R) = 0 \quad \text{for every } L \in \mathcal{L}, R \in \mathcal{R}, \text{ and every } c \text{ with } \nu(c) > 0.$$

Robust coarse gluing rules out cancellations between different fibers and is the local condition used here for boundary completeness.

3 the covariance obstruction

Proposition 1 (coarse gluing obstruction). *For every admissible pair $L \in \mathcal{L}$ and $R \in \mathcal{R}$,*

$$\sum_{y \in \mathcal{B}} L(y)R(y)\mu(y) - \sum_{c \in \mathcal{C}} \bar{L}(c)\bar{R}(c)\nu(c) = \sum_{c \in \mathcal{C}} \text{Cov}_{\mu_c}(L, R)\nu(c),$$

where

$$\text{Cov}_{\mu_c}(L, R) := \sum_{y \in \pi^{-1}(c)} L(y)R(y)\mu_c(y) - \bar{L}(c)\bar{R}(c).$$

Thus pairwise coarse gluing succeeds exactly when the displayed weighted sum of conditional covariances is zero. Robust coarse gluing succeeds exactly when the conditional covariance vanishes in every non-null fiber for every admissible pair.

Proof. The finite disintegration identity is

$$\sum_{y \in \mathcal{B}} L(y)R(y)\mu(y) = \sum_{c \in \mathcal{C}} \left(\sum_{y \in \pi^{-1}(c)} L(y)R(y)\mu_c(y) \right) \nu(c).$$

Subtracting

$$\sum_{c \in \mathcal{C}} \left(\sum_{y \in \pi^{-1}(c)} L(y)\mu_c(y) \right) \left(\sum_{y \in \pi^{-1}(c)} R(y)\mu_c(y) \right) \nu(c)$$

gives precisely the stated sum of conditional covariances. The final sentence is the definition of robust coarse gluing. $\square$

Remark 1 (integrated cancellation). *Pairwise equality can occur through cancellation between fibers. That may be enough for a specific admissible pair (L, R) , but it is not a robust local composition principle. Robust gluing requires the fiberwise condition, because an internal cut should not rely on unrelated positive and negative hidden-fiber errors canceling elsewhere. For example, if all the relevant conditional covariances have the same sign, then pairwise equality forces each of them to vanish; without such a sign condition, cancellations can hide an incomplete boundary.*

Corollary 1 (diagnosis of incomplete coarse data). *Assume the complete interface variable y is known to give the correct gluing pairing for the chosen admissible class. If a coarse proposal $c = \pi(y)$ fails robust coarse gluing, meaning that some admissible pair has nonzero conditional covariance in a non-null fiber, then c is not complete for that admissible class. The lost information lies in distinctions inside the fiber $\pi^{-1}(c)$ that affect the joint product LR .*

The corollary does not claim that every failed gluing test in physics means missing variables. If no finer complete model is known, the failure might instead mean that the cut is inadmissible or that the proposed region weight is not the object that composes. The conclusion here is conditional: given a complete interface model, failure after projection diagnoses loss of relevant fiber information.

4 factor-generated predictive completion

The covariance obstruction identifies failure. The next step is to name the completion forced by the admissible factors. The word “predictive” is used in a specific sense: if the right or exterior factors are the prediction functionals relevant to the cut, then the completion preserves exactly the information those factors need for gluing. It is not claimed to be a predictive-state representation in the full controlled-dynamical sense of [17].

Definition 4 (factor-generated predictive completion). *Let $\pi : \mathcal{B} \rightarrow \mathcal{C}$ be a coarse proposal. Define an equivalence relation $\sim$ on $\mathcal{B}$ by*

$$y \sim y' \iff \pi(y) = \pi(y'), \quad L(y) = L(y') \text{ for all } L \in \mathcal{L}, \quad R(y) = R(y') \text{ for all } R \in \mathcal{R}.$$

The quotient

$$\widehat{\mathcal{C}} := \mathcal{B} / \sim$$

with quotient map $\widehat{\pi} : \mathcal{B} \rightarrow \widehat{\mathcal{C}}$ is the factor-generated predictive completion of the coarse proposal. It is a refinement of $\mathcal{C}$. The induced map

$$\kappa : \widehat{\mathcal{C}} \rightarrow \mathcal{C}, \quad \kappa(\widehat{\pi}(y)) = \pi(y),$$

is well-defined.

The completion keeps exactly the distinctions in the fiber that any admissible factor can see.

This differs from two nearby notions. Kemeny-Snell lumpability asks when a state projection preserves Markov evolution for a transition kernel [16]; here the quotient is generated by the left and right factors relevant to one compositional cut. Predictive-state representations encode state by tests of future observations [17]; here the only retained tests are the admissible factors needed for gluing region weights.

Theorem 1 (minimal finite completion). *Let $\widehat{\mu} = \widehat{\pi}_* \mu$. Then:*

1. *Every admissible factor $L \in \mathcal{L}$ and $R \in \mathcal{R}$ is constant on equivalence classes and therefore descends to a function $\widehat{L}, \widehat{R} : \widehat{\mathcal{C}} \rightarrow [0, \infty)$.*
2. *For every admissible pair,*

$$\sum_{y \in \mathcal{B}} L(y) R(y) \mu(y) = \sum_{\widehat{c} \in \widehat{\mathcal{C}}} \widehat{L}(\widehat{c}) \widehat{R}(\widehat{c}) \widehat{\mu}(\widehat{c}).$$

3. *$\widehat{\mathcal{C}}$ is the coarsest refinement of $\mathcal{C}$ through which all admissible factors descend. More explicitly, if $q : \mathcal{B} \rightarrow Q$ is any other quotient such that π , every $L \in \mathcal{L}$, and every $R \in \mathcal{R}$ descend through q , then there is a map $h : Q \rightarrow \widehat{\mathcal{C}}$ satisfying*

$$\widehat{\pi} = h \circ q.$$

Proof. If $y \sim y'$, all admissible factors have the same values at y and y' , so each factor is constant on $\sim$ -classes and descends to the quotient. This proves the first claim.

For the second claim, write $[y]$ for the equivalence class of y . Since L and R are constant on each class,

$$\sum_{y \in \mathcal{B}} L(y)R(y)\mu(y) = \sum_{\hat{c} \in \hat{\mathcal{C}}} \hat{L}(\hat{c})\hat{R}(\hat{c}) \sum_{y: \hat{\pi}(y)=\hat{c}} \mu(y),$$

which is the stated formula with the pushforward measure $\hat{\mu}$.

For minimality, suppose $q: \mathcal{B} \rightarrow Q$ makes π , all left factors, and all right factors descend. Descent means, for example, that each $L \in \mathcal{L}$ has the form $L = \tilde{L} \circ q$, and therefore is constant on the fibers of q . If $q(y) = q(y')$, then $\pi(y) = \pi(y')$ and all admissible factor values agree, hence $y \sim y'$. Therefore the equivalence classes of q refine the equivalence classes of $\sim$. The assignment $h(q(y)) = \hat{\pi}(y)$ is well-defined and gives $\hat{\pi} = h \circ q$. $\square$

Remark 2 (relative minimality). *The completion is minimal only relative to the chosen admissible factor families. Enlarging the class of regions, observations, or boundary perturbations may refine $\hat{\mathcal{C}}$. This relativity is not a defect. It is what makes boundary completeness a testable property of a modelling class rather than an absolute ontology.*

Proposition 2 (finite completion algorithm). *Assume, in addition, that the admissible families are finite:*

$$\mathcal{L} = \{L_1, \dots, L_m\}, \quad \mathcal{R} = \{R_1, \dots, R_n\}.$$

For each $y \in \mathcal{B}$, form the signature

$$s(y) := (\pi(y), L_1(y), \dots, L_m(y), R_1(y), \dots, R_n(y)).$$

Then the factor-generated predictive completion is obtained by partitioning $\mathcal{B}$ into the level sets of s . Equivalently, two complete interface values are identified exactly when they have the same coarse label and the same values for every admissible left and right factor.

Proof. The level-set relation $s(y) = s(y')$ is precisely the equivalence relation in Definition 4, because the universal quantifiers over $\mathcal{L}$ and $\mathcal{R}$ are exactly the finite lists $\{L_1, \dots, L_m\}$ and $\{R_1, \dots, R_n\}$ in this case. Therefore its quotient is $\hat{\mathcal{C}} = \mathcal{B}/\sim$, and the minimality statement is exactly Theorem 1. $\square$

This gives a constructive finite test. Given a proposed coarse boundary, one computes which hidden fiber distinctions are still seen by the admissible left and right factors. Those distinctions, and only those distinctions, are the finite completion variables required by the chosen modelling class.

Remark 3 (standard-Borel analogue). *When $\mathcal{B}$ and $\mathcal{C}$ are standard-Borel spaces and π admits regular conditional probabilities, Proposition 1 has the same formal expression with sums replaced by integrals. A full measurable completion theorem would replace the finite quotient by the sigma-algebra generated by π and by all functions $L \in \mathcal{L}$ and $R \in \mathcal{R}$, together with an appropriate measurable factorization statement. That extension is not needed for the finite examples proved here.*

5 sharp-limit synchronization gap

The same missing-variable phenomenon appears in sharp positive limits. This section is the variational shadow of the finite gluing theorem, not a separate foundation.

Let the complete interface variable be $y = (c, e)$, where c is the proposed coarse boundary value and e is a hidden completion variable. Suppose that left and right factors have large-deviation forms

$$L_a^\varepsilon(c, e) \asymp \exp[-W_1(a, c, e)/\varepsilon], \quad R_d^\varepsilon(c, e) \asymp \exp[-W_2(c, e, d)/\varepsilon],$$

where ε is a sharpness parameter such as a noise strength, and $A^\varepsilon \asymp \exp[-W/\varepsilon]$ means $-\varepsilon \log A^\varepsilon = W + o(1)$. Under standard Laplace-principle hypotheses [10, 11], complete gluing gives a generator of the form

$$W(a, d) = \inf_{c, e} [W_1(a, c, e) + W_2(c, e, d) + I(c, e)],$$

where $I(c, e)$ is the large-deviation rate contributed by the interface measure; in a uniform finite interface model this term may be zero. The following proposition itself is only an inequality about functions with finite infima.

Proposition 3 (synchronization gap). *For fixed a, c, d , assume the following infima are finite:*

$$\inf_e W_1(a, c, e), \quad \inf_e W_2(c, e, d), \quad \inf_e [W_1(a, c, e) + W_2(c, e, d)].$$

Define

$$\Delta(a, c, d) := \inf_e [W_1(a, c, e) + W_2(c, e, d)] - \left[\inf_e W_1(a, c, e) + \inf_e W_2(c, e, d) \right].$$

Then $\Delta(a, c, d) \geq 0$. Equality holds whenever the one-sided optimizations admit a common minimizing value of e . A positive gap is a sharp-limit diagnostic that e is a missing boundary-completion variable.

Proof. For every e ,

$$W_1(a, c, e) + W_2(c, e, d) \geq \inf_{e'} W_1(a, c, e') + \inf_{e''} W_2(c, e'', d).$$

Taking the infimum over e on the left gives the inequality. If both one-sided infima are attained at the same e , then evaluating the left side at that common optimizer gives equality. $\square$

The sharp-limit inequality is the standard variational fact that an internal variable must be shared by the two pieces, not optimized independently on each side. The contribution here is the diagnostic interpretation. Missing shared variables can already be seen before the sharp limit as failed gluing over a coarse boundary.

Example 1 (quadratic synchronization gap). *Let the hidden completion variable be $e \in \mathbb{R}$, and suppress the external variables. The constants α and β encode how the left and right preferred hidden values depend on those external variables. Take*

$$W_1(e) = (e - \alpha)^2, \quad W_2(e) = (e - \beta)^2.$$

If the two sides are optimized separately, the two infima are both zero:

$$\inf_e W_1(e) + \inf_e W_2(e) = 0.$$

If the same hidden value e must be shared across the cut, then

$$2(e - \alpha) + 2(e - \beta) = 0, \quad e_* = \frac{\alpha + \beta}{2},$$

and hence

$$\inf_e [(e - \alpha)^2 + (e - \beta)^2] = \frac{(\alpha - \beta)^2}{2}.$$

The gap is positive unless $\alpha = \beta$. In this elementary example, each side has a preferred hidden completion value; gluing through the coarse boundary forces them to synchronize on one shared value.

6 finite-memory stochastic systems

The variational example above shows hidden-variable mismatch in a sharp limit; the finite-memory example shows the same obstruction as an ordinary probabilistic screening failure.

The simplest example is a finite-memory stochastic process. Let $(X_n)_{n \in \mathbb{Z}}$ be a finite-state k -th order Markov chain, so

$$P(x_{n+1} \mid x_0, \dots, x_n) = P(x_{n+1} \mid x_n, x_{n-1}, \dots, x_{n-k+1}).$$

Cut a path at time n . If the proposed boundary is only $c = x_n$, the future transition may still depend on the omitted variables $(x_{n-1}, \dots, x_{n-k+1})$. The complete boundary for arbitrary future gluing is the block

$$y_n = (x_{n-k+1}, \dots, x_n).$$

Proposition 4 (finite-memory completion). *For a finite-state k -th order Markov chain, the block*

$$y_n = (x_{n-k+1}, \dots, x_n)$$

is complete for gluing arbitrary future transition factors. If a projection of y_n omits a coordinate and there exist two positive-probability blocks in the same projected fiber for which that coordinate changes a future transition probability, then the projected boundary fails the gluing test for a suitable future observation.

Proof. The first statement is the standard conversion of a k -th order chain into a first-order Markov chain on length- k blocks. Given y_n , the conditional law of the future depends on the past only through y_n . In the language of this paper, admissible future factors descend to the block boundary, and the complete gluing pairing of Definition 1 is the model's finite sum over the retained block state.

For the second statement, suppose two positive-probability blocks y and y' have the same projected value c but give different transition probabilities for some future symbol z :

$$P(X_{n+1} = z \mid y) \neq P(X_{n+1} = z \mid y').$$

Let the right factor be the likelihood of the event $X_{n+1} = z$. Then the right factor is not constant on the fiber $\pi^{-1}(c)$, so it cannot descend to c . Choosing a left factor that weights y and y' differently produces a nonzero conditional covariance inside the fiber. Proposition 1 then gives the failed gluing test. $\square$

Example 2 (binary second-order memory). *Let $X_n \in \{0, 1\}$ be a second-order Markov chain. The complete boundary at time n is $y = (X_{n-1}, X_n)$. Suppose one proposes the coarse boundary $c = X_n$. In the fiber $c = 0$, there are two hidden states:*

$$y_0 = (0, 0), \quad y_1 = (1, 0).$$

Assume both have positive conditional probability in this fiber and, for simplicity, take $\mu_0(y_0) = \mu_0(y_1) = 1/2$. Let the future event be $X_{n+1} = 1$, and suppose

$$P(X_{n+1} = 1 \mid y_0) = p, \quad P(X_{n+1} = 1 \mid y_1) = q, \quad p \neq q.$$

The corresponding right factor has $R(y_0) = p$ and $R(y_1) = q$. Choose a left factor that distinguishes the two hidden histories, for instance

$$L(y_0) = 1, \quad L(y_1) = 0.$$

Then

$$\bar{L} = \frac{1}{2}, \quad \bar{R} = \frac{p+q}{2}, \quad \mathbb{E}_{\mu_0}[LR] = \frac{p}{2},$$

so

$$\text{Cov}_{\mu_0}(L, R) = \frac{p}{2} - \frac{1}{2} \frac{p+q}{2} = \frac{p-q}{4}.$$

Thus gluing through $c = X_n$ fails whenever the omitted memory coordinate changes the future transition probability and the left side can weight the two hidden histories differently. The missing completion variable is the previous state X_{n-1} .

The indicator left factor used in the example is admissible because the formal class only requires non-negativity; no normalization of individual left or right factors is assumed.

This example connects boundary completion with familiar state-augmentation logic. The compositional point is that the retained boundary must screen the two sides of the cut for the chosen class of future factors. This is close in spirit to causal-state and predictive-state ideas [9, 17], while the formal criterion here is gluing of region weights rather than prediction in a controlled dynamical system.

7 finite gauge-frame model

The stochastic example and the gauge-frame example have the same structure. In both cases, a coarse label discards a hidden variable that one side may still need in order to coordinate with the other. The stochastic hidden variable is memory. The gauge-like hidden variable is a shared boundary frame.

Let G be a finite group, interpreted as a boundary frame group. Let the complete interface variable be $g \in G$, with normalized counting measure. Let $L : G \rightarrow [0, \infty)$ and $R : G \rightarrow [0, \infty)$ be left and right subregion factors. Finite-group gauge models are standard toy settings for isolating gauge-composition questions [8]; here the model is used only as a one-fiber gluing test. Complete gluing uses the shared frame:

$$K_{\text{complete}}(L, R) = \frac{1}{|G|} \sum_{g \in G} L(g) R(g).$$

If one quotients away the shared frame before gluing and keeps only the single orbit label $*$, the coarse product rule gives

$$K_{\text{coarse}}(L, R) = \left(\frac{1}{|G|} \sum_{g \in G} L(g) \right) \left(\frac{1}{|G|} \sum_{g \in G} R(g) \right).$$

The difference is

$$K_{\text{complete}}(L, R) - K_{\text{coarse}}(L, R) = \text{Cov}_G(L, R),$$

the covariance over the group fiber.

Proposition 5 (finite edge-frame completion). *In the finite gauge-frame model, quotienting out the shared boundary frame before gluing preserves the same gluing result as the complete interface for all admissible factors only if the covariance over the group fiber vanishes for all admissible pairs. Otherwise the frame g , or an equivalent relative-frame variable, is a required completion variable for local composition.*

Proof. This is Proposition 1 with $\mathcal{B} = G$, $\mathcal{C} = \{*\}$, and μ the normalized counting measure. The single coarse fiber is all of G , so the conditional covariance is exactly the group covariance displayed above. $\square$

Example 3. Let $G = \mathbb{Z}_2 = \{0, 1\}$, let $L(0) = R(0) = 1$, and let $L(1) = R(1) = 0$. Then

$$K_{\text{complete}} = \frac{1}{2}, \quad K_{\text{coarse}} = \frac{1}{4}.$$

The frame value is not observable as an external label after quotienting, but it is still needed to align the two local factors. Removing it before gluing changes the composed weight.

This is the precise sense in which the toy model anticipates edge modes. The claim is not that continuum gauge-theory edge modes have been derived. The claim is that a compositional theory should expect extra boundary variables whenever quotienting local frame data destroys the correlations needed for gluing. In field theory the corresponding variables may be boundary frames, charges, fluxes, or corner data, and their treatment requires the additional geometric machinery developed in the gauge and covariant phase-space literature [3, 4, 6, 5].

8 discussion

The results above show how a proposed boundary variable can fail a local composition test in a controlled finite setting. The covariance obstruction identifies hidden fiber dependence. The factor-generated completion gives the coarsest refinement, relative to an admissible factor class, through which the local factors descend. The sharp-limit synchronization gap gives the corresponding variational diagnostic.

The construction is relative. A boundary can be complete for one class of regions and observations while incomplete for another. This is the same kind of relativity that appears in sufficient statistics and predictive state representations: the sufficient state depends on which predictions or losses are being considered. The compositional version asks which interface data are sufficient for gluing region weights.

The paper does not use maximum-caliber or Schrödinger-bridge machinery. Those subjects are nearby because they also organize path weights and conditioning, but the finite theorem here uses only factor descent and covariance over interface fibers.

Several limitations should be kept explicit. First, the finite minimality theorem concerns factor descent. Pairwise covariance cancellations can produce accidental equality without making the coarse variable a robust local boundary. Second, the sharp-limit proposition is an elementary variational inequality, not a full theorem about all large-deviation limits. Third, the finite gauge-frame model is only a toy model. Continuum gauge theory and gravity require constraints, boundary conditions, presymplectic reduction, and possibly boundary extensions. The finite result supplies a diagnostic logic, not the full field-theoretic construction.

These limitations are also what make the result useful. The paper provides a test: specify the admissible local factors, choose a proposed boundary variable, and check whether gluing survives projection. When it does not, the failure is not merely a numerical error. It identifies the distinctions inside the discarded fiber that the two sides still use jointly.

9 conclusion

Complete boundary data are the variables through which local region weights compose. If a complete interface is coarse-grained, failed gluing has a specific obstruction: conditional covariance between

the left and right subregion factors inside the hidden fibers. The missing data can be organized as a factor-generated predictive completion, and in sharp limits the same omission appears as a synchronization gap between one-sided optimizers.

Finite-memory stochastic processes show the mechanism in familiar form: memory variables that affect future transitions must be retained at the cut. The finite gauge-frame model shows the same logic in a gauge-like setting: a shared frame may be redundant before cutting but necessary for composing the cut pieces. In the finite compositional setting, this gives a precise finite analogue of the slogan that edge modes are boundary-completion variables forced by local composition.

References

- [1] R. Rosemarin, “Cut-Objective Predictive Mechanics: Boundary response geometry from composable region weights,” manuscript, 2026.
- [2] R. P. Feynman and A. R. Hibbs, *Quantum Mechanics and Path Integrals*, McGraw-Hill, New York, 1965.
- [3] W. Donnelly and L. Freidel, “Local subsystems in gauge theory and gravity,” *Journal of High Energy Physics* **2016**, 102 (2016), arXiv:1601.04744 [hep-th].
- [4] D. Harlow and J. Wu, “Covariant phase space with boundaries,” *Journal of High Energy Physics* **2020**, 146 (2020), arXiv:1906.08616 [hep-th].
- [5] V. Iyer and R. M. Wald, “Some properties of Noether charge and a proposal for dynamical black hole entropy,” *Physical Review D* **50**, 846–864 (1994), arXiv:gr-qc/9403028 [gr-qc].
- [6] J. Lee and R. M. Wald, “Local symmetries and constraints,” *Journal of Mathematical Physics* **31**, 725–743 (1990).
- [7] J. C. Baez, B. Fong, and B. S. Pollard, “A compositional framework for Markov processes,” *Journal of Mathematical Physics* **57**, 033301 (2016), arXiv:1508.06448 [math-ph].
- [8] R. Dijkgraaf and E. Witten, “Topological gauge theories and group cohomology,” *Communications in Mathematical Physics* **129**, 393–429 (1990).
- [9] C. R. Shalizi and J. P. Crutchfield, “Computational mechanics: Pattern and prediction, structure and simplicity,” *Journal of Statistical Physics* **104**, 817–879 (2001), arXiv:cond-mat/9907176 [cond-mat.stat-mech].
- [10] H. Touchette, “The large deviation approach to statistical mechanics,” *Physics Reports* **478**, 1–69 (2009), arXiv:0804.0327 [cond-mat.stat-mech].
- [11] P. Dupuis and R. S. Ellis, *A Weak Convergence Approach to the Theory of Large Deviations*, Wiley, New York, 1997.
- [12] R. Oeckl, “General boundary quantum field theory: Foundations and probability interpretation,” *Advances in Theoretical and Mathematical Physics* **12**, 319–352 (2008), arXiv:hep-th/0509122 [hep-th].
- [13] R. A. Fisher, “On the mathematical foundations of theoretical statistics,” *Philosophical Transactions of the Royal Society of London. Series A* **222**, 309–368 (1922).

- [14] D. Blackwell, “Equivalent comparisons of experiments,” *The Annals of Mathematical Statistics* **24**, 265–272 (1953).
- [15] E. L. Lehmann and G. Casella, *Theory of Point Estimation*, 2nd ed., Springer, New York, 1998.
- [16] J. G. Kemeny and J. L. Snell, *Finite Markov Chains*, D. Van Nostrand, Princeton, 1960.
- [17] M. L. Littman, R. S. Sutton, and S. Singh, “Predictive representations of state,” *Advances in Neural Information Processing Systems* **14**, 1555–1561 (2001).
- [18] T. Fritz, “A synthetic approach to Markov kernels, conditional independence and theorems on sufficient statistics,” *Advances in Mathematics* **370**, 107239 (2020), arXiv:1908.07021 [math.ST].